

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME:CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:20

Annexure A MCQ TEST

Criteria	Conceptual Understanding(3)	Accuracy of Answer(2.5)	Application of Knowledge(2)	Completeness of Response(1.5)	Clarity & Presentation(1)	Total(10)
Weightage	30%	25%	20%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						

Code of Conduct:

I J.Raju(192325117) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

J.Raju

RUBRICS FOR CALCULATION

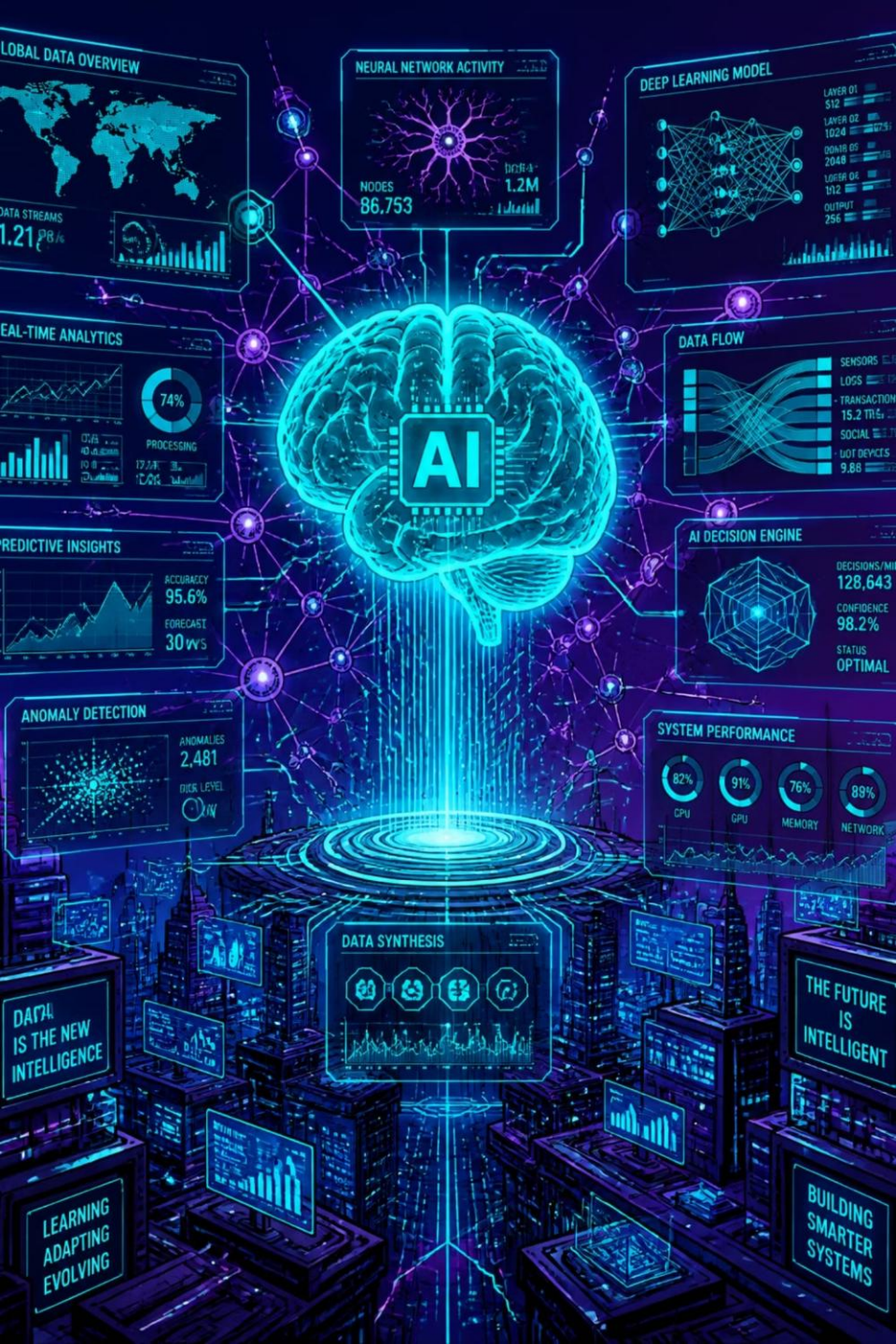
Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand



J.Raju

192325117

CSA1607 Data Warehousing And DataMining



DATA WAREHOUSING & BUSINESS INTELLIGENCE

Inside a Business Brain

How OLAP Turns Millions of Records into Smart Decisions

Imagine managing **500 supermarkets** with millions of daily transactions. How do managers instantly discover which product sells best, in which city, during which season? OLAP is the intelligence engine behind those answers.

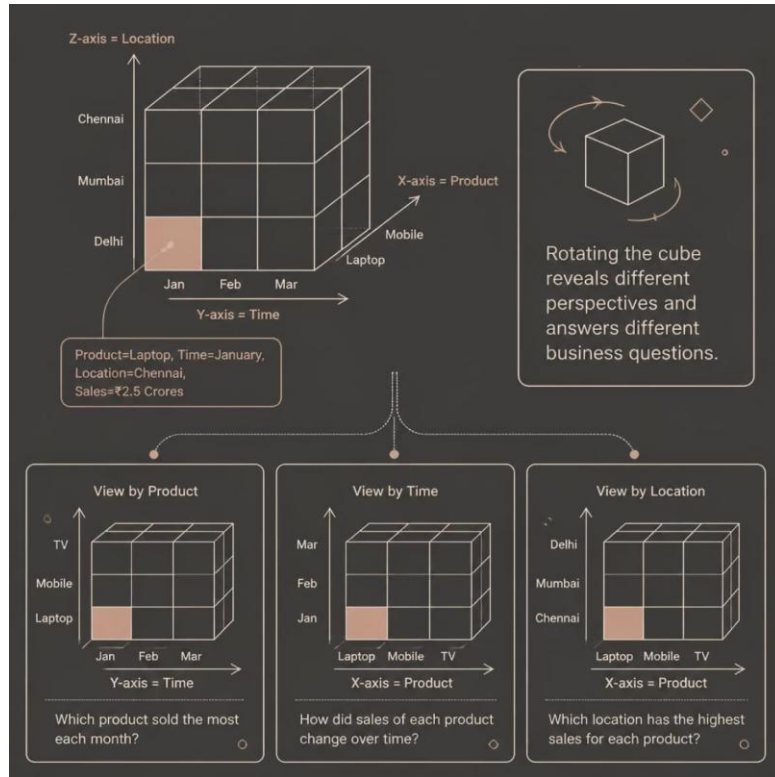
What Is OLAP, Really?

Think of OLAP as Google Maps for business data — it lets you zoom in, pan out, and explore from any angle, instantly.



OLAP (Online Analytical Processing) transforms raw historical data into multi-dimensional insights — in seconds, not hours.

The OLAP Data Cube



One Cube. Hundreds of Questions.

The same dataset answers completely different questions depending on your perspective:

📄 **Example:** Laptop · January · Chennai = ₹2.5 Crores

By Product

Which laptop model sold most in Chennai this January?

By Time

Did sales peak in Q1 or Q4?

By Location

Which city generated the highest revenue?

Four OLAP Operations

These four operations let analysts navigate the data cube with surgical precision — like controls on a business intelligence cockpit.



Slice

Isolate one dimension — e.g., show **only Chennai** data across all products and months.



Dice

Apply multiple filters — e.g., **Laptops in Chennai** during **January–March**.



Drill Down

Go deeper — **Year** → **Month** → **Day**. Uncover granular trends hidden in aggregates.



Roll Up

Summarize upward — **Day** → **Month** → **Year**. Get the big-picture view instantly.

OLAP in the Real World

From retail floors to hospital corridors, OLAP powers the dashboards that drive critical decisions every day.



Retail & E-Commerce

Which products to restock?
Which city leads revenue?
OLAP answers in real time.



Banking & Finance

Track loan performance, detect
fraud patterns, and analyze
profitability by branch.



Healthcare

Monitor patient outcomes,
optimize resource allocation,
and predict seasonal demand.



Manufacturing

Identify production
bottlenecks, track supply chain
efficiency, and forecast
demand.

THE ROAD AHEAD

The Future of OLAP

OLAP is evolving — converging with AI, cloud platforms, and predictive models to power the next generation of intelligent systems.



AI & Machine Learning

OLAP feeds clean, structured data into ML pipelines for smarter predictions.



Cloud Data Warehouses

Snowflake, BigQuery, and Redshift embed OLAP at massive scale.



Predictive Analytics

From hindsight to foresight — OLAP powers proactive business intelligence.

"Data is the new oil. OLAP is the refinery that transforms it into intelligent decisions."

Thank you — Questions? 🙋